



CBIR Literature Review

👤 Created by	👤 Yashas Shetty
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Introduction

Medical imaging has experienced exponential growth in both volume and variety, creating an urgent need for efficient retrieval systems that can support clinical decision-making, medical education, and research. Content-Based Image Retrieval (CBIR) systems address this challenge by enabling similarity-based searches using visual features such as texture, shape, and color, rather than relying solely on text-based annotations [1], [2]. Unlike general-purpose CBIR systems, medical CBIR must meet stringent requirements for accuracy, reliability, and clinical relevance, making robust evaluation methodologies essential [3].

The evaluation of medical CBIR systems presents unique challenges. Medical images span diverse modalities (X-ray, CT, MRI, ultrasound, histopathology), anatomical regions, and pathological conditions, each requiring specialized feature extraction and similarity measures [4]. Furthermore, the clinical utility of a retrieval system depends not only on technical performance metrics but also on its ability to support diagnostic workflows and educational applications [5]. This review synthesizes current evaluation methodologies, examining the metrics, datasets, and protocols that have emerged as standards in the field.

Methodology

Content-Based Image Retrieval (CBIR) in medical imaging enables the retrieval of clinically similar images based on visual characteristics rather than textual annotations. The proposed methodology follows a structured pipeline consisting of dataset preparation, preprocessing, feature extraction, similarity computation, and performance evaluation. This framework is widely adopted in medical CBIR research because it supports scalable retrieval from large diagnostic image repositories and assists clinical decision-making through visual comparison [1][2].

The first stage involves dataset preparation and preprocessing of medical images. Images are collected from curated clinical datasets or institutional archives and standardized to ensure consistency across samples. Preprocessing typically includes resizing images to a uniform resolution, normalizing pixel intensities, and removing noise using filtering techniques. For oral cancer detection tasks, region-of-interest extraction is particularly important to isolate lesion areas from surrounding tissue, improving retrieval relevance and reducing background interference [3]. These preprocessing steps enhance feature stability and improve retrieval performance across heterogeneous medical imaging modalities [1].

Following preprocessing, discriminative visual features are extracted to represent each image numerically. Traditional CBIR systems rely on handcrafted descriptors such as texture features derived from Gray Level Co-occurrence Matrices (GLCM), Local Binary Patterns (LBP), and intensity histograms, which capture spatial and structural characteristics of tissue patterns [4]. However, modern medical CBIR systems increasingly employ deep learning models to obtain hierarchical feature representations. Convolutional Neural Networks (CNNs), implemented using frameworks such as PyTorch, automatically learn complex diagnostic features from medical images and have demonstrated superior performance in disease-specific retrieval tasks, including cancer imaging [2][5]. In practical implementations, feature vectors are extracted from intermediate CNN layers and stored as compact representations of image content.

After feature extraction, images are indexed in a feature database to enable efficient retrieval. Each image is represented as a fixed-length feature vector, and dimensionality reduction techniques such as Principal Component Analysis may be applied to reduce computational complexity while preserving discriminative information. During retrieval, the feature vector of a query image is compared with stored vectors using similarity measures such as Euclidean

distance or cosine similarity. Images are ranked according to similarity scores, and the top-k most similar images are returned as retrieval results. This similarity-based ranking allows clinicians or researchers to compare new cases with previously observed patterns and supports visual diagnostic assistance [4][5].

The performance of the CBIR system is evaluated using standard information retrieval metrics. Precision, recall, and mean average precision (mAP) are commonly used to quantify retrieval accuracy and ranking quality. In medical applications, evaluation may also involve expert validation to assess clinical relevance of retrieved images. High retrieval accuracy indicates that the feature representation effectively captures diagnostic characteristics of lesions and supports reliable image-based comparison [1][2]. This evaluation framework ensures that the CBIR system provides meaningful support for medical image analysis and decision-making.

Standard Evaluation Metrics

Precision and Recall

Precision and recall constitute the foundational metrics for evaluating CBIR system performance. Precision measures the proportion of retrieved images that are relevant to the query, while recall quantifies the proportion of all relevant images in the database that are successfully retrieved [6], [7]. These metrics provide complementary perspectives on retrieval effectiveness: high precision indicates few false positives, while high recall indicates comprehensive coverage of relevant images [8].

Mean Average Precision (MAP)

Mean Average Precision (MAP) has emerged as a preferred single-value metric that combines the impact of precision and recall across multiple queries [9], [10]. MAP computes the average precision at each relevant document position in the ranked retrieval results, providing a comprehensive assessment of ranking quality. Popova et al. demonstrated a 14.49% improvement in MAP using a combined features approach across 57 medical image categories from the IRMA database [10].

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